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Vector Databases for RAG: An Introduction ×

Today's goals
Complete any 3 learning items - 3/3
○ Complete a module
Complete a practice item

Today's Skill Points 80 XP
See skill progress

Module 1
Introduction to Vector Databases and Chroma DB

Welcome to the Course
- ✓ IBM Product Spotlight: watsonx.data
  Reading • 2 min
- ✓ Video: Course Introduction
  Video • 3 min
- ✓ Course Overview
  Reading • 5 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- ✓ Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

Introduction to Vector Databases and Similarity Search
- ✓ Vector Database Concepts
  Video • 6 min
- ✓ Reading: Vector Databases Versus Traditional Databases
  Ungraded Plugin • 5 min
- ✓ Vector Database Types
  Video • 9 min
- ✓ Applications of Vector Databases
  Video • 6 min
- ✓ Reading: Similarity Search
  Ungraded Plugin • 30 min
- ✓ Lab: Similarity Search by Hand
  Ungraded App Item • 30 min
- ✓ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Introduction to Vector Databases and Similarity Search
  Practice Assignment • Grade: 100%

Exploring Chroma DB
- ✓ Chroma DB Key Concepts and Architecture
  Video • 6 min
- ✓ Reading: Chroma DB Filtering
  Ungraded Plugin • 15 min
- ✓ Reading: Similarity Search and HNSW in Chroma DB
  Ungraded Plugin • 30 min
- ✓ Lab: Similarity Search on Text Using Chroma DB and Python
  Ungraded App Item • 30 min
- ✓ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Exploring Chroma DB
  Practice Assignment • Grade: 100%

Module Summary and Evaluation
- ✓ Reading: Summary and Highlights: Introduction to Vector Databases and Chroma DB
  Reading • 3 min
- ✓ Reading: Cheat Sheet: Introduction to Vector Databases and Chroma DB
  Ungraded Plugin • 15 min
- ✓ Graded Quiz: Introduction to Vector Databases and Chroma DB
  Graded Assignment • Grade: 100%
- ✓ [Optional] Discussion Prompt: Meet and Greet
  Discussion Prompt • 10 min

Module 2
Vector Databases for Recommendation Systems and RAG

Chroma DB Database Operations
- ✓ Essential Database Operations in Chroma DB
  Video • 5 min
- ✓ Lab: Similarity Search on Employee Records using Python and Chroma DB
  Ungraded App Item • 1h
- ✓ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Chroma DB Database Operations
  Practice Assignment • Grade: 100%

Develop a Recommendation System and Connect Learned Concepts to RAG
- ✓ How Vector Databases Power RAG
  Video • 6 min
- ○ Lab: Practice Project: Food Recommendation System Using Chroma DB
  Ungraded App Item • 2h
- ✓ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Develop a Recommendation System and Connect Learned Concepts to RAG
  Practice Assignment • Grade: 80%

Module Summary and Evaluation
- ○ **Summary and Highlights: Vector Databases for Recommendation Systems and RAG**
  Reading • 2 min
- ○ Reading: Vector Databases for Recommendation Systems and RAG Cheat Sheet
  Ungraded Plugin • 15 min
- ○ Graded Assignment: Graded Quiz: Vector Databases for Recommendation Systems and RAG
  Graded Assignment • 21 min

Course Wrap Up
- ○ What's Next: Explore watsonx.data
  Reading • 1 min
- ○ Course Wrap-Up
  Video • 4 min
- ○ Congratulations and Next Steps
  Reading • 2 min
- ○ Team and Acknowledgments
  Reading • 2 min

Summary and Highlights: Vector Databases for Recommendation Systems and RAG

6 XP • Advanced RAG System Implementation   3 XP • Applied Generative AI Development and Strategy

0:08 / 2:00   1x

Congratulations! You have completed this lesson. At this point in the course, you know:

- That Chroma DB operations form the foundation for building intelligent, vector-based applications
- That collections are a way to organize your data within Chroma DB
- That you define your embedding model using Chroma DB's embedding_functions
- That you create a collection using the create_collection method
- That metadata helps you keep track of the purpose and contents of your collections
- That you connect to an existing collection by using the get_collection method
- That you alter an existing collection by using the modify method
- That you use the add method to insert documents into a collection
- That you get data from a collection using the get method
- That you update existing data in a collection by using the update method
- That you delete data from a collection using the delete method
- That you use the HNSW space configuration parameter to specify the distance function when performing approximate nearest neighbor searches in Chroma DB
- How to analyze data with similarity search
- That RAG enhances LLM response quality by retrieving relevant external information, helping the model generate more accurate and well-supported outputs
- That vector databases are the foundation that makes Retrieval-Augmented Generation work
- That the tasks vector databases can perform in a RAG pipeline include: embedding source documents and user prompts, storing embeddings, retrieving most relevant matches, and providing the retrieved content for prompt augmentation
- That using a vector database for all relevant RAG steps helps prevent critical mistakes, speeds up application development, and optimizes performance
- That some RAG pipeline tasks, such as chunking, advanced retrieval logic, prompt augmentation, and LLM integration, usually happen outside the database
- That RAG frameworks, such as LangChain and LlamaIndex, can wrap around your vector database and help manage the full RAG pipeline, simplifying RAG application development even further
- How to build a recommendation system using Chroma DB

Mark as completed

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